

From Polycystic Ovary Syndrome to Polyendocrine Metabolic Ovarian Syndrome: Rationale and Clinical Implications

*Sangeeta Tiwari**

Abstract

Background: Polycystic ovary syndrome (PCOS), recently renamed polyendocrine metabolic ovarian syndrome (PMOS), is a common, heterogeneous disorder with reproductive, endocrine, metabolic and psychological consequences. The traditional term PCOS has been criticized because it overemphasizes ovarian morphology and suggests the presence of pathological cysts, although the ultrasound finding usually represents multiple arrested follicles and is neither universal nor essential for diagnosis.

Objective: To summarize rationale for replacing PCOS with PMOS and to describe the expected implications of terminology change for diagnosis, patient communication, multidisciplinary care, research and public health messaging.

Methods: This narrative review was prepared from the recent global consensus on naming of PCOS/PMOS and current international guidance on assessment and management of condition.

Summary: The term PMOS better reflects multisystem nature of the disorder. “Polyendocrine” acknowledges involvement of multiple hormonal pathways, including androgen excess, insulin-related pathways, neuroendocrine regulation and ovarian function. “Metabolic” highlights central cardiometabolic risks, including insulin resistance, dyslipidemia, type 2 diabetes risk, obesity-associated complications and cardiovascular risk factors. Retaining “ovarian” recognizes that ovulatory dysfunction and ovarian morphology remain clinically important.

Conclusion: Adoption of PMOS is more than a semantic change. It supports clearer patient education, earlier recognition in patients without polycystic ovarian morphology, and more integrated care across gynecology, endocrinology, primary care, nutrition, psychology and fertility services.

Keywords: polycystic ovary syndrome; polyendocrine metabolic ovarian syndrome; PMOS; PCOS; insulin resistance; hyperandrogenism; metabolic syndrome; women’s health

Introduction:

In the past, polycystic ovarian syndrome has been classified as a gynecological or reproductive disorder. Clinical experience and current data, however, indicate that the illness affects much more than only the ovary. Hyperandrogenism, ovulatory dysfunction, insulin resistance, cardiometabolic risk factors, and psychosocial strain are frequently included. In order to bring terminology into line with the biology and clinical realities of the disorder,

a global consensus procedure was used to suggest the new word, polyendocrine metabolic ovarian syndrome (PMOS).

As disease names influence patient comprehension, professional attitudes, diagnostic paths, funding goals, and research conceptualization, the change is therapeutically significant. Patients and medical professionals may believe that ovarian cysts are necessary for diagnosis or that the illness only affects fertility if the name is based around “cysts.” This misunderstanding could cause care to be fragmented and diagnosis to be delayed.

* Principal, Amaltas Institute of Nursing Sciences, Amaltas University, Dewas, M.P., India
Corresponding Author: Sangeeta Tiwari, tiwarisangeeta94@gmail.com

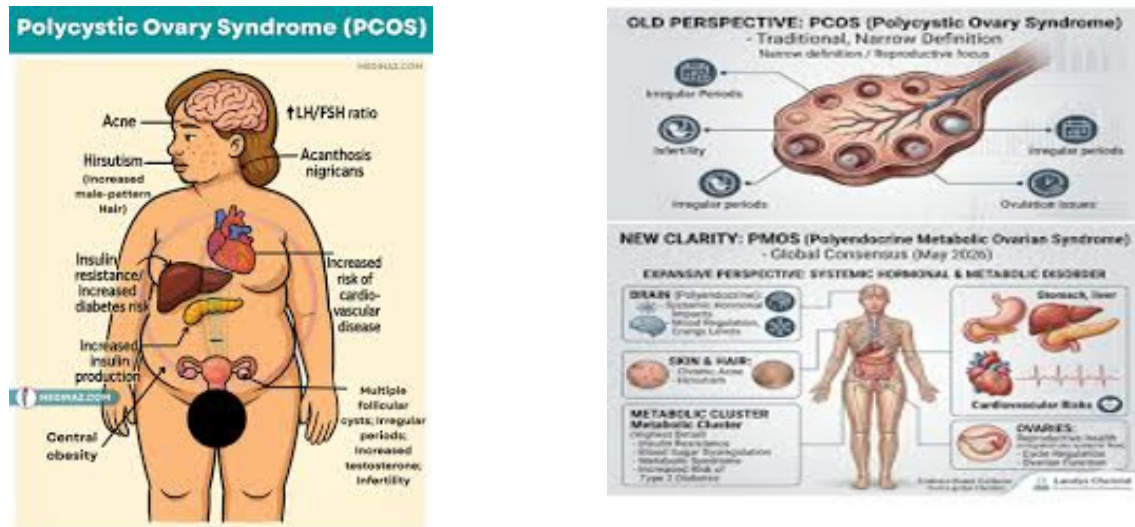


Fig 1. A and B: Clinical manifestations of Polycystic Ovary Syndrome

Why the term PCOS was considered inaccurate:

The word “polycystic” is misleading. The ovarian appearance classically described in PCOS usually reflects multiple small, arrested follicles rather than true pathological ovarian cysts. In addition, many affected individuals do not have polycystic ovarian morphology on ultrasound, and some may meet diagnostic criteria through hyperandrogenism and ovulatory dysfunction alone.

The term “ovary syndrome” also underrepresents endocrine and metabolic dimensions of disorder. Ovarian

dysfunction is important, but it is part of a broader network of hormonal and metabolic disturbance. A name that appears purely gynecological may reduce attention to insulin resistance, obesity-associated risk, dyslipidemia, hypertension, type 2 diabetes risk, cardiovascular risk factors, fatty liver risk and mental health concerns.

Meaning of the new term PMOS:

The consensus name polyendocrine metabolic ovarian syndrome retains the familiar syndrome structure while making the disease framework more accurate. Each component of the name has clinical meaning.

Table 1. Interpretation of the term Polyendocrine Metabolic Ovarian Syndrome

Component	Clinical interpretation	Practical implication
Polyendocrine	Involvement of multiple hormonal pathways, including androgen excess, insulin-related pathways, neuroendocrine function and ovarian hormones.	Encourages assessment beyond pelvic ultrasound and menstrual history.
Metabolic	Recognition of insulin resistance and long-term cardiometabolic risk.	Supports screening for weight-related risk, glucose intolerance, dyslipidemia, blood pressure and lifestyle needs.
Ovarian	Ovarian dysfunction remains part of the condition through ovulatory disturbance, menstrual irregularity and ovarian morphology in some patients.	Preserves the reproductive and fertility dimensions of the disorder.
Syndrome	A cluster of related signs and symptoms rather than a single-organ disease.	Supports individualized, multidisciplinary care.

Clinical features and pathophysiological emphasis:

PMOS commonly presents with irregular menstrual cycles, anovulation or oligo-ovulation, clinical or biochemical hyperandrogenism, acne, hirsutism, scalp hair thinning, subfertility, weight gain or difficulty with weight management, and metabolic abnormalities. The clinical phenotype is heterogeneous; therefore, absence

of one feature, such as polycystic ovarian morphology, should not exclude the diagnosis when other criteria are fulfilled.

The metabolic dimension is central. Insulin resistance can amplify androgen production and contribute to weight gain, dysglycemia and cardiometabolic risk. Psychological features, including distress related to body image, acne, hirsutism, infertility, delayed diagnosis and stigma, should also be recognized as part of holistic care.

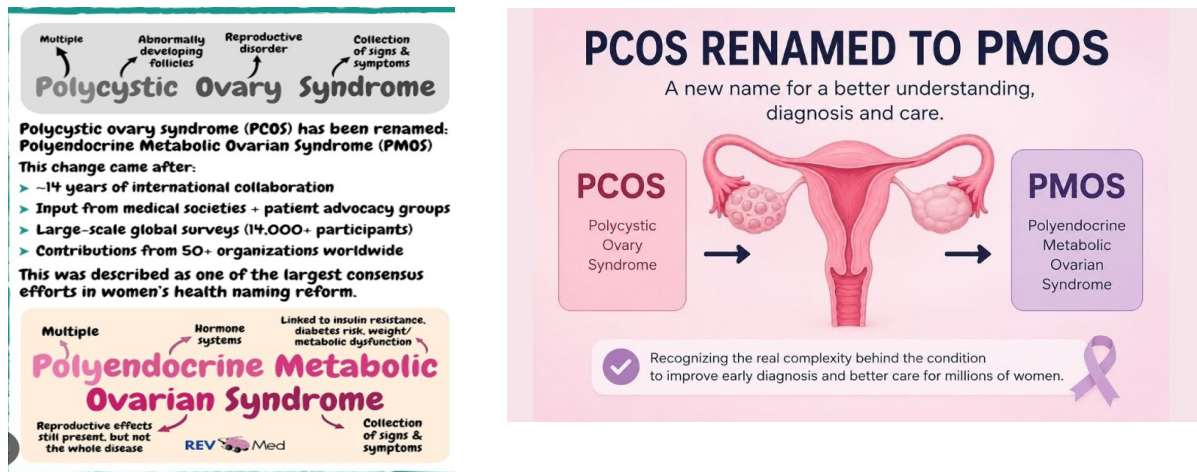


Fig.2: From Polycystic Ovary Syndrome to Polyendocrine Metabolic Ovarian Syndrome

Implications for diagnosis:

The name change does not mean that PMOS is a new disease. It is the updated terminology for the condition previously known as PCOS. Diagnostic criteria and clinical evaluation should continue to follow current evidence-based guidance while terminology transitions in clinical records, patient education materials and professional communication.

In adults, contemporary guidance supports diagnosis using revised evidence-based criteria: two of the following after exclusion of other causes—clinical or biochemical hyperandrogenism, ovulatory dysfunction, and polycystic ovarian morphology on ultrasound or elevated anti-Müllerian hormone where appropriate. When both irregular cycles and hyperandrogenism are present, ultrasound or anti-Müllerian hormone testing may not be required. Adolescents require more cautious diagnostic assessment because normal pubertal physiology can overlap with PMOS features.

Benefits for patient communication:

Patient comprehension may be enhanced by a more precise name. According to PMOS, the condition is more

than just “cysts in the ovary,” and symptoms like irregular periods, acne, hirsutism, weight problems, insulin resistance, exhaustion, and psychological discomfort can all be related. By characterizing the condition as a systemic endocrine-metabolic ailment rather than one that is only related to reproduction, the name may also lessen stigma associated with fertility.

During the transition period, clinicians may use both terms, such as “PMOS, previously known as PCOS,” to maintain continuity for patients, prescriptions, insurance documentation, coding systems and educational resources.

Implications for multidisciplinary care:

PMOS requires coordinated care. Gynecologists remain central for menstrual, ovarian and fertility issues, but comprehensive management may also involve endocrinologists, primary care physicians, dietitians, dermatologists, psychologists, diabetes educators and fertility specialists. The terminology supports this broader approach by making endocrine and metabolic care visible in the name itself.

Routine assessment should consider menstrual pattern,

signs of androgen excess, fertility goals, metabolic risk, body mass index and waist measures where appropriate, blood pressure, glycemic status, lipid profile, sleep and psychological wellbeing. Management should be individualized according to age, symptoms, reproductive plans, cardiometabolic risk and patient priorities.

Public health and research implications:

The global burden of PMOS is substantial. The condition affects a large proportion of women and people assigned female at birth during reproductive years, and its long-term implications extend into diabetes prevention, cardiovascular risk reduction, fertility care and mental health. A terminology shift may improve public health messaging and promote research that examines the whole condition rather than isolated ovarian morphology.

For researchers, PMOS may encourage study designs that integrate reproductive, metabolic, endocrine and psychosocial outcomes. For educators and policymakers, it can support more accurate training materials and broader service models.

Implementation challenges:

It takes time for medical terminology to evolve. Updating textbooks, recommendations, electronic medical records, diagnostic codes, public health resources, patient advocacy content, and exam curricula are among the challenges. Due of PCOS's strong linkages to patient communities and clinical practice, dual terminology is probably going to persist for a while.

In order to prevent confusion, clinicians should clarify that PMOS is an updated term meant to enhance accuracy and care rather than a distinct diagnosis from PCOS. Simple explanations should be used in patient-facing materials, and it should be emphasized that care is still determined by each patient's unique symptoms and risks.

Conclusion:

The renaming of PCOS to PMOS represents a major conceptual shift from an ovary-centered label to a multisystem endocrine-metabolic framework. The new terminology better captures the central roles of hormonal dysregulation, insulin resistance and cardiometabolic risk while preserving the importance of ovarian dysfunction. Adoption of PMOS may improve diagnosis, reduce stigma, broaden multidisciplinary care and align patient education with contemporary evidence.

Acknowledgement:

The author acknowledges the use of the original draft material as the basis for this revised narrative review.

Declarations:

Funding: None

Conflicts of interest: None

References:

1. Teede HJ, Bahri Khomami M, Morman R, et al. Polyendocrine metabolic ovarian syndrome, the new name for polycystic ovary syndrome: a multistep global consensus process. *Lancet*. 2026; S0140-6736(26)00717-8.
2. Endocrine Society. Polyendocrine Metabolic Ovarian Syndrome: New name to improve diagnosis and care of condition affecting 170 million women worldwide. News release. 12 May 2026.
3. Teede HJ, Tay CT, Laven JJE, et al. Recommendations from the 2023 International Evidence-based Guideline for the Assessment and Management of Polycystic Ovary Syndrome. *J Clin Endocrinol Metab*. 2023;108(10):2447-2469.
4. International Evidence-based Guideline for the Assessment and Management of Polycystic Ovary Syndrome 2023. Monash University, Centre for Research Excellence in Women's Health in Reproductive Life.